higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENTATION TECHNOLOGY
NQF LEVEL 4

27 FEBRUARY 2019

This marking guideline consists of 7 pages.

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SECTION A

QUESTION 1

1.1	1.1.1	C		
	1.1.2	A		
	1.1.3	B		
	1.1.4	C		
	1.1.5	B		
			(5 × 2)	(10)
1.2	1.2.1	E		
	1.2.2	D		
	1.2.3	G		
	1.2.4	A		
	1.2.5	B		
			(5 × 1)	(5)
1.3	1.3.1	False		
	1.3.2	False		
	1.3.3	True		
	1.3.4	True		
	1.3.5	True		
			(5 × 1)	(5)
1.4	1.4.1	Turbidity		
	1.4.2	Interlock		
	1.4.3	pH meter		
	1.4.4	PLC		
	1.4.5	DCS		
			(5 × 2)	(10)
				[30]

TOTAL SECTION A: 30

SECTION B

QUESTION 2

2.1 A hydrostatic level sensor is used to measure the pressure created by a liquid.✓ The greater the height of the liquid, the greater the pressure.✓ The sensor must be close to the bottom of the tank✓ and either fitted on the outside via a thread or a flange,✓ or entered into the top of the tank✓ and then attached to the bottom.✓ The pressure exerted by the liquid creates the movement of a flexible piece of metal or ceramic known as a diaphragm.✓ The position of the diaphragm is measured✓ and a signal is produced that relates to the level of liquid in the tank.✓ (9)

2.2
- Strain-gauge load cell
- Piezoelectric load cell
- Hydraulic load cell
- Pneumatic load cell (4)

2.3
- Normal installation: The transmitter is mounted levelled with a minimum-level tap.
- Elevated-zero installation: The transmitter is mounted above the high-pressure tap.
- Suppressed-zero installation: The transmitter is mounted below the high-pressure tap. (3 × 2) (6)

2.4
- Source of gamma radiation
- Continuous detector
- Microprocessor (3)

2.5 In general the gamma source emitting radiations is located externally to the process vessel.✓ These radiations pass through the vessel walls and the material gets accumulated towards the detector.✓ In case when the vessel is empty✓ the transmitted rays arrive at the detector unaffected.✓ As soon as the level of the content increases in the vessel,✓ there will be a decrease in the number of rays reaching the detector.✓ The gamma energy drops in an inverse proportion to the process level.✓ A computer processes the detector signal and transmits the process variable as a 4-20 mA analogue signal.✓ (8)

[30]

QUESTION 3

3.1
- Incoming raw materials may be sampled to ensure compliance with required chemical specifications✓ or to optimise the yield✓ and minimise production costs.✓
- In-line process analysis is normally necessary at various stages to ensure that there will not be excessive waste at the end of the production process✓ owing to failure to meet the final product specification.✓
- Chemical composition analysis is also employed to assure public health and safety,✓ as in the case of municipal water treatment and sewage disposal plants.✓
- The objective in all cases is to determine which chemical elements are present✓ or how much of one or more of several known elements are present.✓ (9)

3.2 3.2.1 Dissolved-oxygen analysers continuously measure the concentration of oxygen by promoting a desired chemical reaction✓ and then measuring the reaction product.✓ The concentration is determined by measuring the change in electrolytic conductance.✓✓ The conductance is proportional to the amount of dissolved oxygen in the sample.✓ (5)

3.2.2
- It is used in high-purity water systems, ✓ such as those used as feed water to high-pressure steam generators. ✓
- It can monitor oxygen removal ✓ which is essential to minimise corrosion in a steam-generating plant. ✓ (2 × 2) (4)

3.3 3.3.1 The dew cell element consists of a thin-walled, hollow-metal socket,✓ wrapped with tape filled with salt crystals (lithium chloride).✓ Two wires are wrapped over the tape✓ and connected to a regulated, alternating-current voltage source.✓ The electric circuit between the wires is completed by salt crystals.✓ (5)

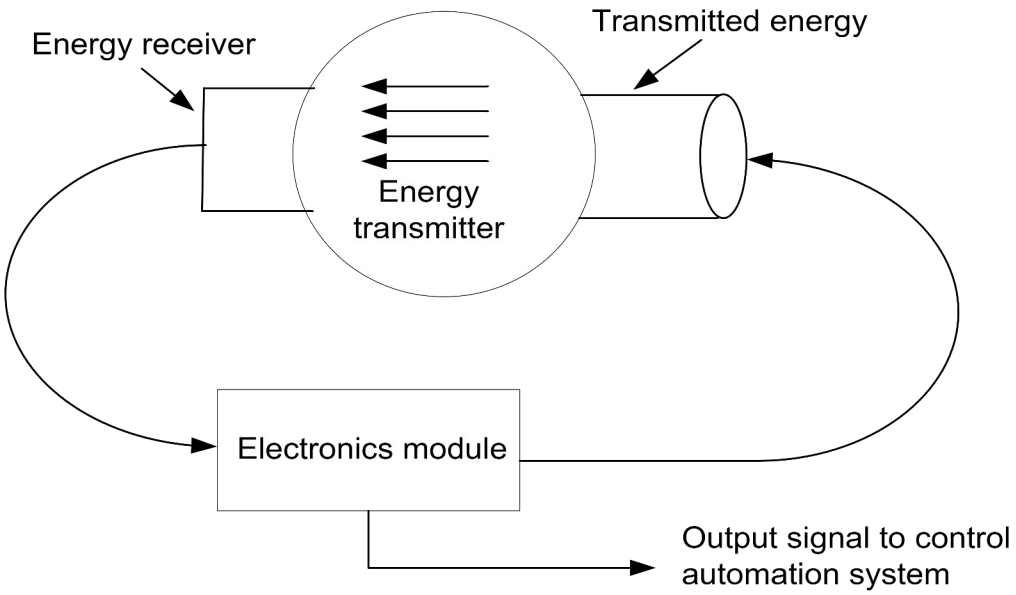
3.3.2 When the sensing element is exposed to the sample atmosphere, water condensing on the crystals✓ forms an ionic solution✓ which allows an electric current to flow between the wires.✓ This current in turn heats the solution and raises its vapour pressure.✓ As more water condenses, more current flows✓ resulting in a further increase of the solution vapour pressure until equilibrium is reached.✓ A temperature sensor inside the hollow socket detects temperature and provides a signal for readout.✓ (7)

[30]

QUESTION 4

4.1
- Scatter effect
- Absorption effect
- Coriolis effect
- Torque effect (4)

4.2

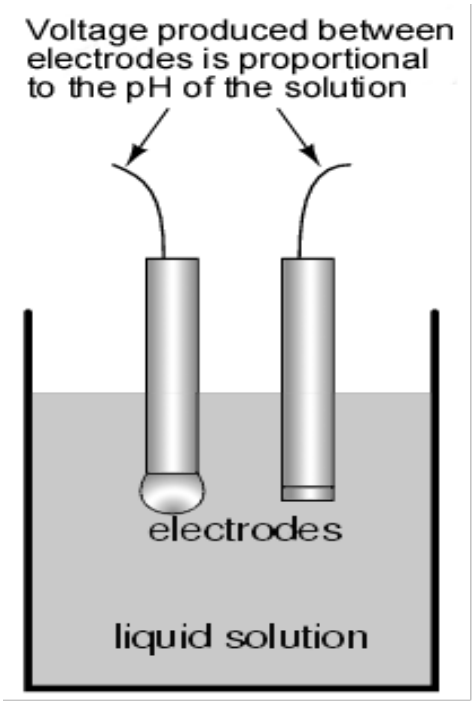


An energy transmitter is located on one side of the process✓ and the energy receiver on the opposite side.✓ As the energy beam strikes the suspended solids,✓ some of the energy is absorbed and reflected.✓ The receiver measures the remaining energy signals✓ and the analyser measures the difference in the transmitted energy✓ versus the amount of the received energy.✓ The difference in the energies is then computed✓ into the amount of suspended solids in the process stream.✓

(FOUR marks for sketch + NINE marks for discussion) (13)

4.3
- Maintains the desired specifications in certain food processes.
- Maintains a specific acid solution or specific basic solution.
- Monitors the level of corrosive chemicals in the liquid. (3)

4.4



A pH measurement loop is essentially a battery✓ where the positive terminal is the measuring electrode✓ and the negative terminal is the reference electrode.✓ The measuring electrode, which is sensitive to the hydrogen ion,✓ develops a potential (voltage) directly related to the hydrogen-ion concentration of the solution.✓ The reference electrode provides a stable potential✓ against which the measuring electrode can be compared.✓ (THREE marks for sketch + SEVEN marks for discussion) (10)

[30]

QUESTION 5

5.1 5.1.1 A status change (alarm condition) of a single high-limit alarm occurs when the input rises above the trip point.✓ The status will return to a non-alarm condition when the input falls below the dead band.✓✓

5.1.2 A status change (alarm condition) of a single low-limit alarm occurs when the input falls below the trip point.✓ The status will return to a non-alarm condition when the input rises above the dead band.✓✓

(2 × 3) (6)

5.2
- The output signal is not linear with the input signal.
- There is a large time lag in terms of error corrections.
- The output varies with the variations of the supply.
- The output is sensitive to vibrations.
- The system has a limited range or distance.
- Leakages and pressure loss is also a problem.

(Any 5 × 1) (5)

5.3

- In an open-loop control system there is no feedback signal to a controller✓ to reset the process, ✓ or to be compared with the controlled signal.✓ Any deviations in the process must be corrected manually.✓

- A closed-loop control system has a feedback signal✓ which is constantly compared with the set point✓ so that the controlled variable is maintained at some desired value.✓ (7)

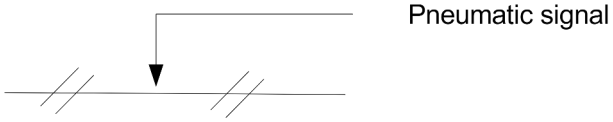
5.4 5.4.1



5.4.2



5.4.3



(3 × 2) (6)

5.5 5.5.1 A control valve is a device that changes the rate of flow of a fluid within a process-control system. (2)

5.5.2 A positioner is a device that is installed between the controller and the valve actuator.✓ It moves the valve stem in accordance with the value of the input signal.✓ This is done by increasing or decreasing the amount of air supply to the diaphragm of the valve✓ and overcomes the valve-stem friction, forces exerted on the plug and changes in the fluid pressure.✓ (4)

[30]

TOTAL SECTION B: 120
GRAND TOTAL: 150

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